

Muhammad Hamza Rao

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Profile

Engineering student with hands-on experience in applied AI, data analysis, and building practical systems using Python. Strong interest in developing AI-driven solutions, working with real-world data, and rapidly prototyping ideas into functional applications.

Technical Skills

Programming: Python, C++, Java, SQL
AI & Machine Learning: PyTorch, Scikit-learn, NLP, Model Evaluation
Data & Analysis: Pandas, NumPy, Data Preprocessing, Feature Engineering
Tools: Git, Jupyter Notebook, Linux
Visualization: Matplotlib, Seaborn, Excel
Concepts: Data-Centric AI, Time-Series Analysis, Problem Solving

Education

Bachelor's Degree in Computer and Electrical Engineering

Technische Hochschule Ingolstadt

B.Eng Autonomous Vehicle Engineering

Expected 03/2027

- Strong academic performance: Machine Learning (1.3), Python (1.7), C++ (1.7)
- Relevant coursework: Machine Learning, Data Structures & Algorithms, Software Engineering, Computer Vision

Experience

Data Science Virtual Intern — British Airways

June 2025

- Built data preprocessing pipelines and performed feature engineering using Python
- Applied machine learning techniques to analyze customer behavior patterns
- Worked with structured datasets to derive insights for business use cases

Projects

LLM-Based Phishing Email Detector — Python, OpenAI API

- Built an AI system to classify phishing emails using prompt-engineered LLM workflows
- Combined rule-based logic with model outputs to improve reliability
- Generated explainable outputs for decision support

DataPainter – Synthetic Dataset Generation Tool — Python, OpenCV

- Built an interactive tool to draw labeled data points and generate datasets in real time
- Converted user input into structured CSV datasets for machine learning workflows
- Enabled controlled experimentation with custom data distributions

Time-Based Chicago Crime Rate Predictive Analysis — Python, Pandas, Time-Series Analysis

- Analyzed time-based crime data to identify patterns and trends across different periods
- Processed and structured datasets for temporal analysis and visualization
- Applied predictive techniques to explore crime rate forecasting and data-driven insights

Research

Seminar Paper: Reinforcement Learning for Real-Time Decision Making

- Researched reinforcement learning methods and real-world applications
- Explained key AI concepts and limitations in structured format

Additional

Certifications

- Introduction to TensorFlow for AI, ML & Deep Learning — DeepLearning.AI
- Python for Data Science, AI & Development — IBM

Languages: English (C1), German (A2-B1)